SAPTHAGIRI NPS UNIVERSITY
UNMATCHED EXCELLENCE, UNLIMITED POTENTIAL

Department of Computer Science & Engineering-Allied Branches
Operating Systems-24BECSE303
AY-2025-26 Sem-III
Question Bank

MODULE-1				
Sl.No	**Question**	**RBT Level**	**COs**	**Marks**
1.	Explain Services and objectives of OS	L2	CO1	10
2.	Explain abstract view of computer system.	L2	CO1	5
3.	Explain System call and its various types in detail **OR** List and describe the six major categories of system calls provided by most operating systems.	L2	CO1	10
4.	The concept of "policy versus mechanism" guide the design of an OS. Justify	L3	CO1	5
5.	How does virtualization impact OS security and performance?	L3	CO1	5
6.	Explain the main challenges in implementing a purely layered operating system?	L2	CO1	8
7.	Explain the core components of an operating system.	L2	CO1	8
8.	Differentiate between the user view and the system view of an operating system.	L3	CO1	5
9.	Define operating systems, Explain dual mode operating system with neat diagram.	L2	CO1	10
10.	Distinguish between multiprogramming and multitasking	L3	CO1	5
11.	Explain briefly process management and memory management	L2	CO1	7
12.	Compare the Monolithic kernel structure with the Microkernel structure.	L3	CO1	10
13.	Explain application program interface with example.	L2	CO1	4
14.	Explain the layered approach to OS design with neat diagram and also mention its advantages and disadvantages.	L2	CO1	7
15.	Explain the concept of a hybrid OS structure, using a modern OS like Windows or macOS as an example.	L2	CO1	7
16.	Define virtual machine? With neat diagram explain concept of virtual machine.	L2	CO1	7
17.	Explain Monolithic kernel structure with a neat diagram.	L2	CO1	10
18.	Explain Microkernel structure with a neat diagram.	L2	CO1	10
MODULE-2				
19.	Explain Inter process communication and its types with a neat diagram.	L2	CO1	5
20.	Describe Process illustrating its transition diagram along with PCB.	L2	CO1	10
21.	Discuss the implementation of Inter process communication using message passing systems in detail.	L2	CO1	10

22.	Define thread by clearly indicating the benefits of multithreading along with various multi-threading models. **OR** Explain Multithreading model with a neat diagram	L2	CO1	7
23.	Discuss the benefits of multithreaded programming.	L2	CO1	5
24.	Describe brief about multithreading concept and explain different multithreading models in operating systems.	L2	CO1	7
25.	Describe the programming challenges of multithreaded programming.	L2	CO1	10
26.	Discuss Inter process communication- shared memory concept with an example.	L2	CO1	10
27.	Define Interprocess Communication (IPC). Why is it required?	L2	CO1	5
28.	Describe direct and indirect communication with respect to message passing mechanism.	L2	CO1	10
29.	Define cascading termination? Explain the parent-child process relationship during process creation.	L2	CO1	10
30.	Describe synchronization in message passing mechanism. **OR** Explain blocking vs non-blocking send/receive message passing mechanism.	L2	CO1	10
31.	Explain process creation in UNIX with fork() and exec() system calls with a neat diagram.	L2	CO1	7
32.	Describe process creation with a neat diagram of a tree of processes on a typical Linux system.	L2	CO1	10
33.	Explain the steps involved in a context switch with an example.	L2	CO1	10
34.	Define scheduling queue? Describe the different types of queues in process scheduling.	L2	CO1	7
35.	Explain the benefits of multithreading models in operating systems. **OR** Compare Many-to-One, One-to-One, and Many-to-Many multithreading models with advantages and disadvantages.	L3	CO1	8
36.	Differentiate between data parallelism and task parallelism with examples.	L3	CO1	7
37.	List and explain different types of parallelism with a neat diagram.	L2	CO1	8